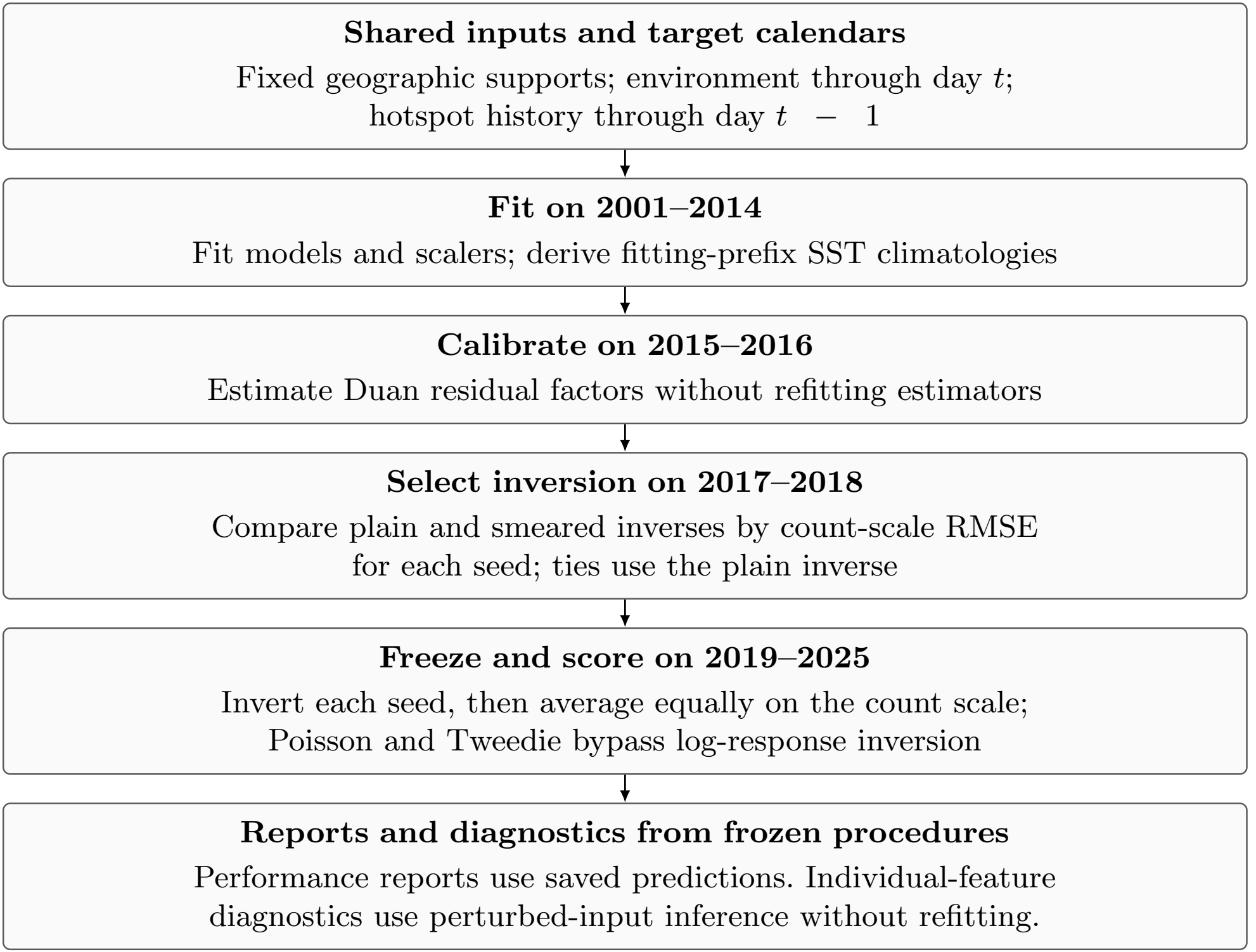
Shared inputs and target calendars
Fixed geographic supports; environment through day t;
hotspot history through day $t - 1$
Fit on 2001–2014
Fit models and scalers; derive fitting-prefix SST climatologies
Calibrate on 2015–2016
Estimate Duan residual factors without refitting estimators
Select inversion on 2017–2018
Compare plain and smeared inverses by count-scale RMSE
for each seed; ties use the plain inverse
Freeze and score on 2019–2025
Invert each seed, then average equally on the count scale;
Poisson and Tweedie bypass log-response inversion
Reports and diagnostics from frozen procedures
Performance reports use saved predictions. Individual-feature
diagnostics use perturbed-input inference without refitting.